DATA CENTER SIMULATION PERFORMANCE REPORT

Workload Profile: FLASH_EVENT | Timestamp: 2026-08-23 18:55:00

OVERALL SLA VERDICT: FAIL (3/4 Checks Satisfied)

1. Experiment Summary & Identification

Parameter	Value
Profile / Workload	flash_event
Execution Duration	600.0 s
Start Time (T1)	1787490898.412288
End Time (T2)	1787491498.412288
Total Requests Executed	7135
Primary Bottleneck	None
Statistical Confidence	HIGH
Failure Classification	SLA_NON_COMPLIANCE, HTTP_REQUEST_FAILURES

2. SLA Scorecard & Compliance Targets

SLA Metric	Measured Value	Target Limit	Result
P95 Response Time	230.0 ms	500.0 ms	PASS
Error Rate	2.55 %	1.0 %	FAIL
Peak CPU Utilization	10.28 %	80.0 %	PASS
Peak Memory Utilization	18.15 %	90.0 %	PASS

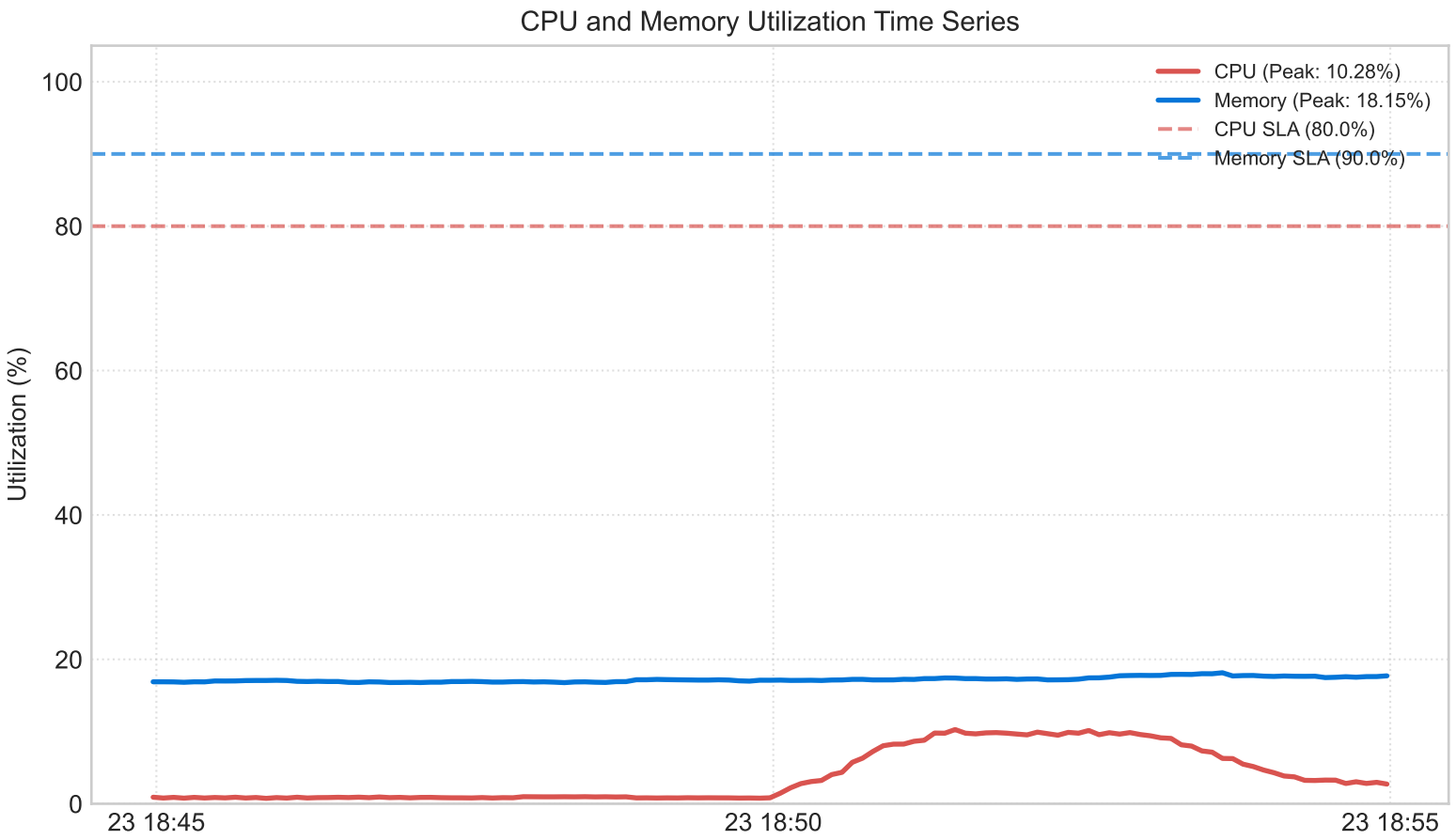
3. Executive Conclusion & Synthesis

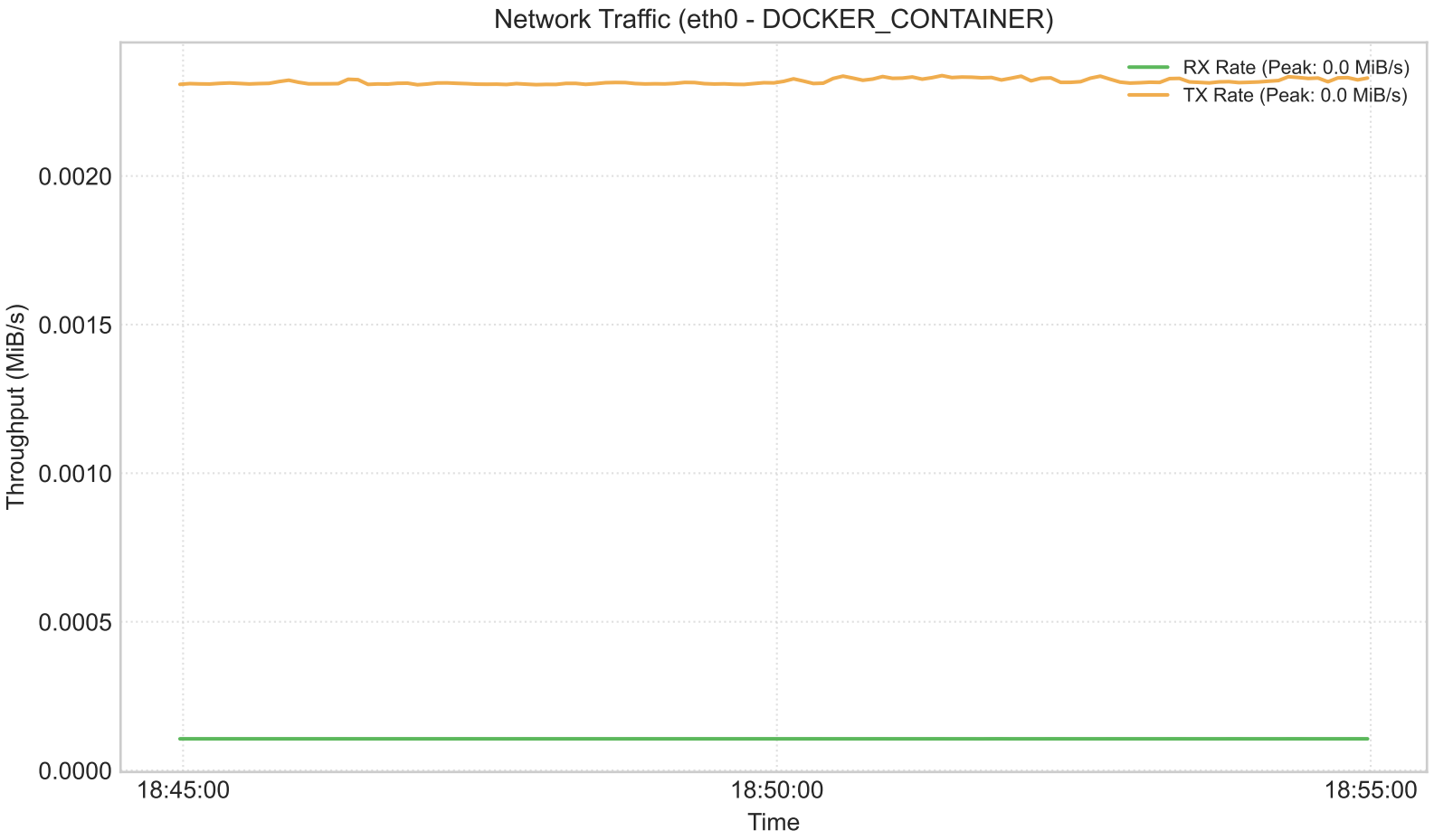
The experiment executed 7135 requests with overall SLA status 'FAIL' (3/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. Primary bottleneck assessed as 'None' (Confidence: HIGH).

4. Diagnostic Observations & Failure Evidence

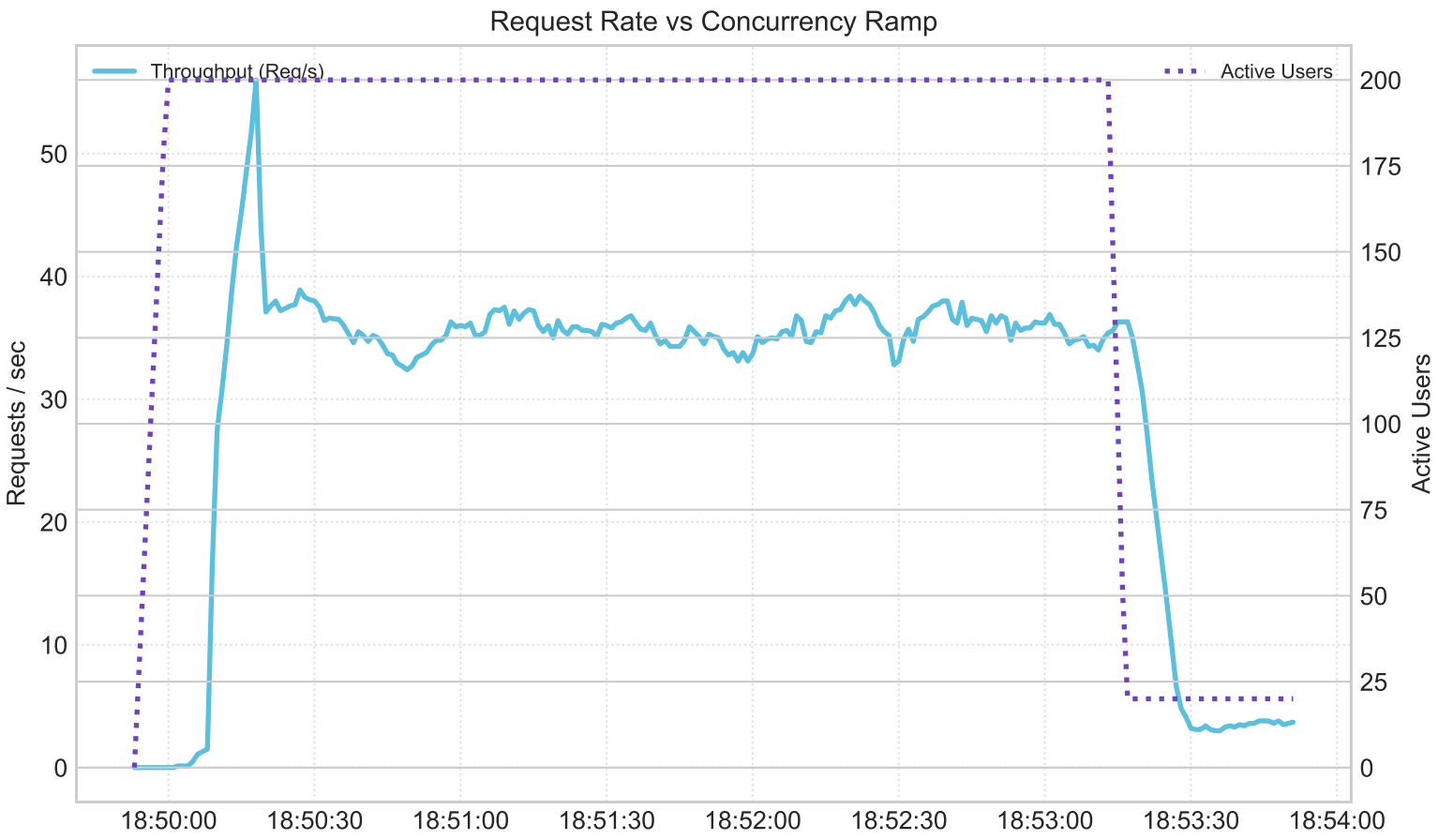
- All measured system and load parameters remained within safe thresholds.
- Locust recorded 182 failed requests (2.55% error rate).
- Failure on POST /login: HTTPError('429 Client Error: Too Many Requests for url: /login') (180 occurrences)
- Failure on GET /health: RemoteDisconnected('Remote end closed connection without response') (2 occurrences)
- SLA Target Violated: Error Rate: measured 2.55 % exceeds limit 1.0 %

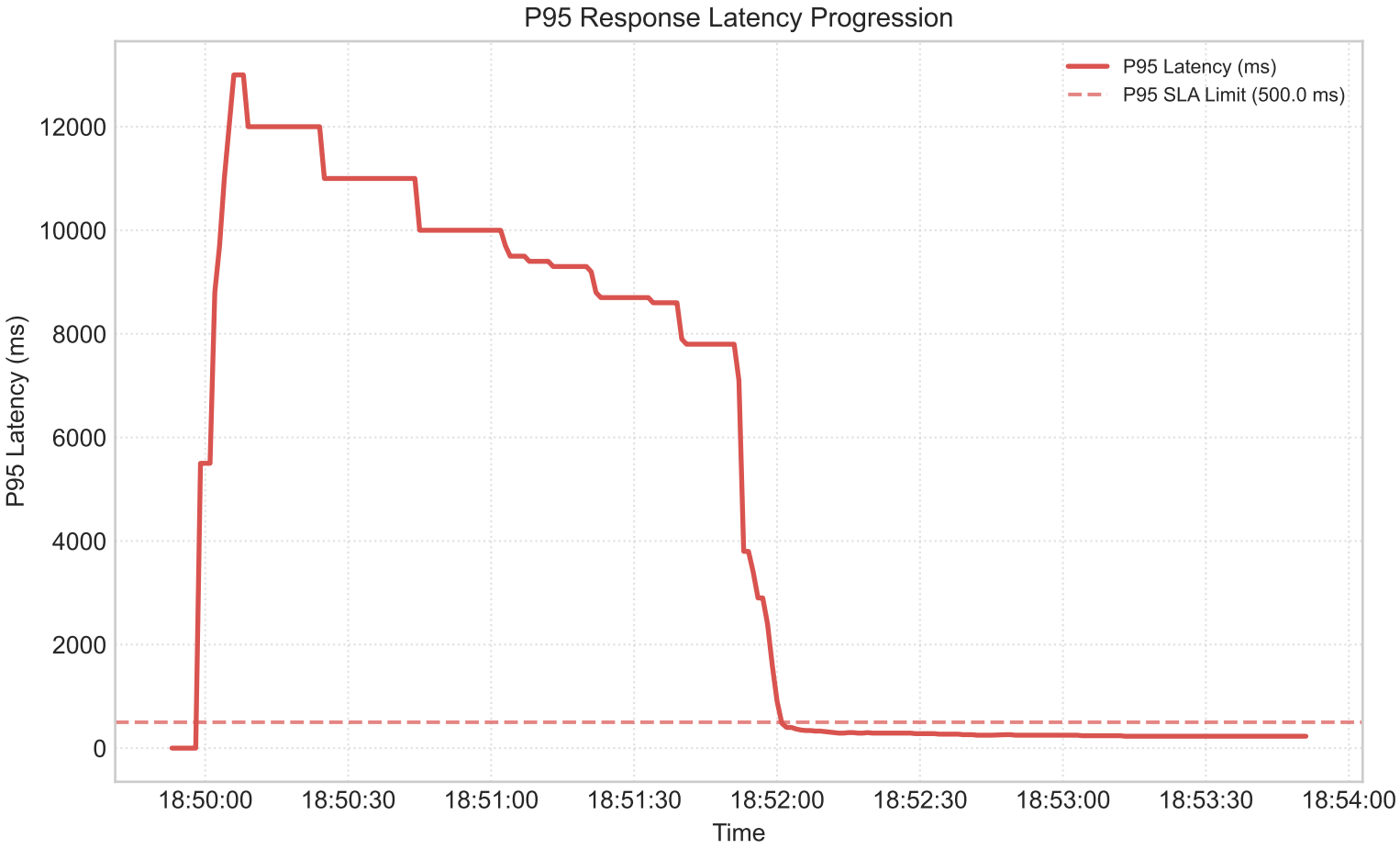
System & Hardware Resource Dynamics





Locust Workload & Latency Profile





Endpoint Breakdown & Chronological Event Timeline

1. HTTP Endpoint Performance Breakdown

Endpoint	Requests	Failures	Avg Latency	P95 Latency	RPS
GET /api/data	1915	0	122.9 ms	230.0 ms	7.97
GET /health	3698	2	42.33 ms	140.0 ms	15.38
POST /login	200	180	10141.11 ms	13000.0 ms	0.83
POST /logout	17	0	117.48 ms	230.0 ms	0.07
GET /me	1122	0	57.37 ms	170.0 ms	4.67
POST /profile/update	183	0	110.22 ms	230.0 ms	0.76

2. Chronological Event Timeline

Time	Event Source & Type	Category	Severity	Description
18:44:58	[RUNNER] EXPERIMENT_START	OBSERVED	INFO	Experiment started at T1 (1787490898.412288)
18:49:53	[LOCUST] LOAD_STARTED	OBSERVED	INFO	Workload generation began with 0 users
18:50:00	[LOCUST] LOCUST_PEAK_USERS	OBSERVED	INFO	Locust concurrency reached peak of 200 active users
18:50:18	[LOCUST] LOCUST_PEAK_RPS	OBSERVED	INFO	Locust throughput reached peak of 56.0 req/s
[PROMETHEUS] NETWORK_PEAK_TX	OBSERVED	INFO	Network transmit rate peaked at 0.0 MiB/s on eth0
[PROMETHEUS] PROMETHEUS_CPU_PEAK	OBSERVED	INFO	CPU utilization peaked at 10.28%
[PROMETHEUS] NETWORK_PEAK_RX	OBSERVED	INFO	Network receive rate peaked at 0.0 MiB/s on eth0
[PROMETHEUS] PROMETHEUS_MEMORY_PEAK	OBSERVED	INFO	Memory utilization peaked at 18.15%
[SLA_ENGINE] SLA_BREACH_DETECTED	INFERRED	CRITICAL	SLA Target Breached: Error Rate was 2.55 % (Limit: 1.0 ...
18:54:58	[RUNNER] EXPERIMENT_END	OBSERVED	INFO	Experiment completed at T2 (1787491498.412288)